

Craig Ssemakula
Cloud Computing
Prof. Azamat Zhamanoz
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Objective

Set up Apache and MySQL servers, create a database, and develop a PHP web application to query and display student information by country.

Implementation

1. XAMPP Installation & Server Setup

Installed XAMPP v3.3.0 and started both Apache (ports 80, 443) and MySQL (port 3306) servers.

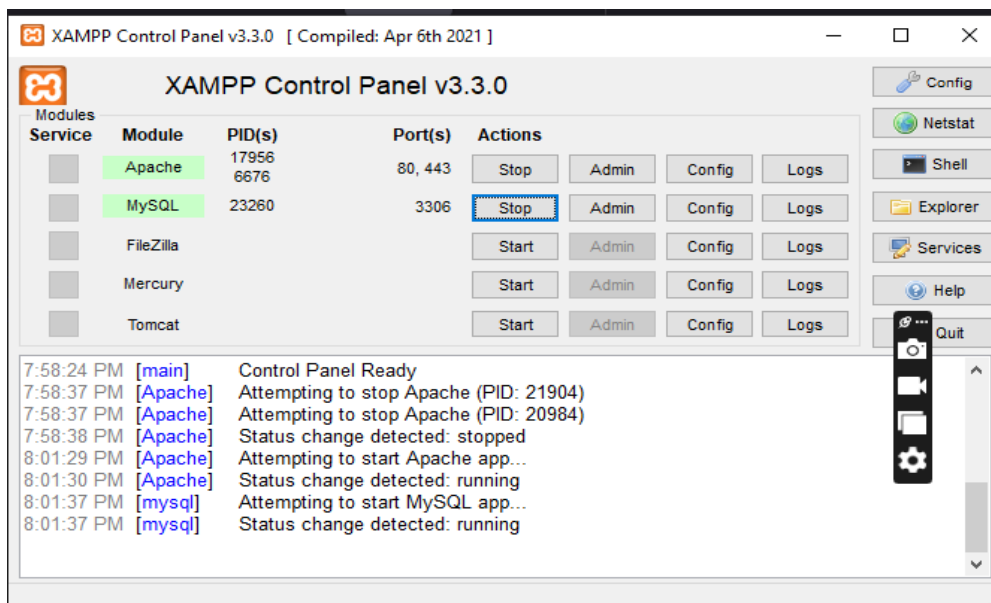


Figure 1: XAMPP Control Panel - Both servers running successfully

2. Database Creation

Accessed phpMyAdmin at <http://localhost/phpmyadmin/> and created school_db_demo database with two tables: countries (10 records) and students (50 records).

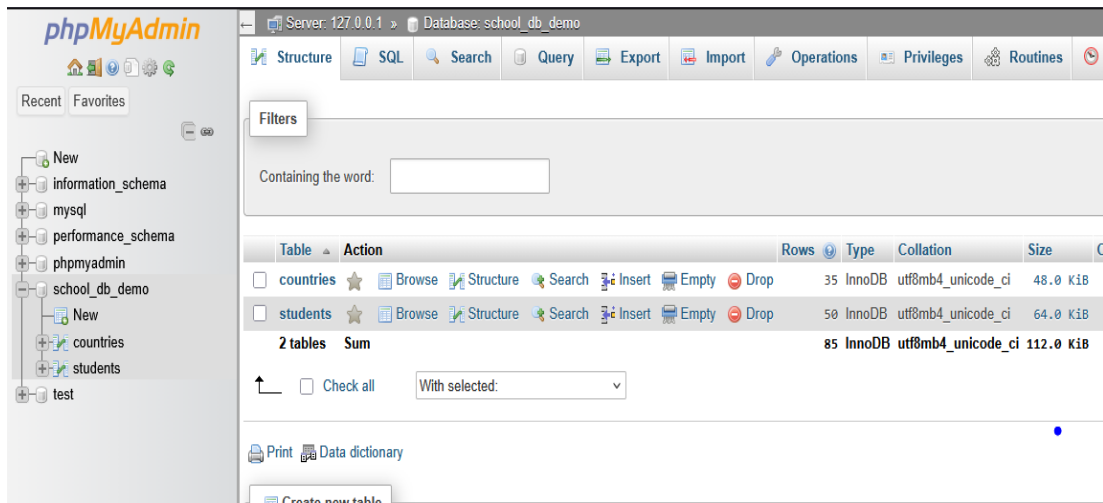


Figure 2: Database structure in phpMyAdmin

	id	country_name	country_code	population
☐ Edit Copy Delete	1	United States	US	33100000
☐ Edit Copy Delete	2	Canada	CA	4000000
☐ Edit Copy Delete	3	Mexico	MX	12900000
☐ Edit Copy Delete	4	Brazil	BR	20300000
☐ Edit Copy Delete	5	Argentina	AR	4600000
☐ Edit Copy Delete	6	United Kingdom	GB	6700000
☐ Edit Copy Delete	7	France	FR	6800000
☐ Edit Copy Delete	8	Germany	DE	8400000
☐ Edit Copy Delete	9	Italy	IT	5900000
☐ Edit Copy Delete	10	Spain	ES	4800000

Figure 3: Countries table with sample data

3. Web Application Setup

Created folder C:\xampp\htdocs\school_db and placed index.php file. The application connects to MySQL and queries student data by country.

Search Students by Country

Country

Brazil

Search

Start typing and pick from the list. (Suggestions come from DB)

Brazil

Code: BR

Population: 203,000,000

Languages: Portuguese

Students (2)

Student Code	Name	Gender	DOB	Email	Phone	Native Language
STU000035	Lucas Santos	M	2004-05-29	lucas.santos35@example.com	+55-21-5550-0135	Portuguese
STU000004	Sofia Silva	F	2003-11-30	sofia.silva4@example.com	+55-11-5550-0104	Portuguese

Figure 4: Web application showing search results for Brazil

Results

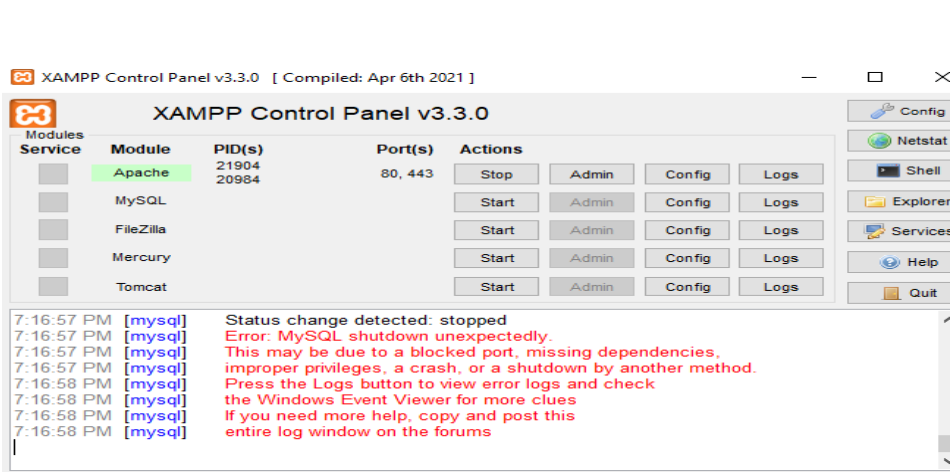
Successfully completed all requirements:

- Apache and MySQL servers running
- Database created with 60 total records
- Web application functioning correctly
- Dynamic search returning accurate results

Troubleshooting

Issue: MySQL failed to start - port 3306 already in use

Solution: Used netstat -ano | findstr :3306 to identify conflicting process and terminated it via Task Manager



Conclusion

Successfully set up a local LAMP development environment using XAMPP.